MATH1141: Higher Mathematics 1A - Algebra

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Contents

Chapter 1

Introduction to Vectors

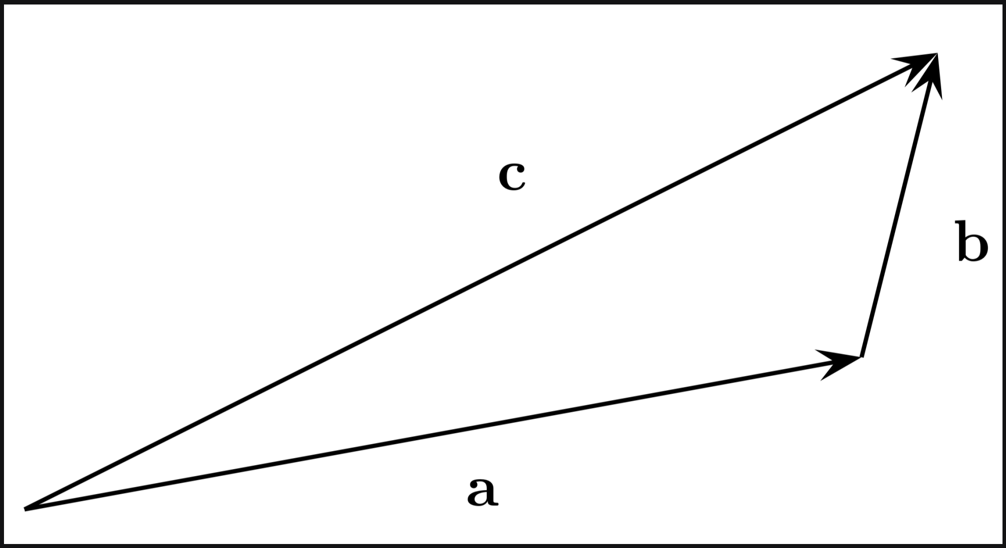
1.1 Vector quantities

- A **scalar** quantity is anything that can be specified by a single number
- A **vector** quantity is specified by both a magnitude and direction

1.1.1 Geometric vectors

- A geometric vector is often represented by an arrow or a directed line segment
- The length of the arrow represents the magnitude of the vector and the arrow points in the direction of the vector
- In these notes, vectors will be denoted as $\mathbf{v}$
- Two vectors are equal if they share the same direction and magnitude. Hence, a translation of a vector would be the same vector.

Vectors $\mathbf{u}$ and $\mathbf{v}$ can be added "head to tail" to produce a new vector $\mathbf{u}+\mathbf{v}$



The zero vector $\mathbf{0}$ has no length and is the only vector with no specific direction. Adding the zero vector to a vector does not change it.

Scalar multiplication is the process of multiplying a vector $\mathbf{v}$ by a scale $\lambda \in \mathbb{R}$. If $\lambda \geq 0$, then $\lambda\mathbf{v}$ denotes the vector with the same direction, but magnitude is scaled by λ, ie. $|\lambda\mathbf{v}| = \lambda|\mathbf{v}|$.

Properties of vector arithmetic

Vector manipulation is defined similarly to regular numbers (the names of these laws should be memorised):

- **Commutative law**: $\mathbf{v}+\mathbf{w}=\mathbf{w}=\mathbf{v}$
- **Additive Associative law**: $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}=\mathbf{w})$
- **Multiplicative Associative law**: $\lambda(\mu\mathbf{v})=(\lambda\mu)\mathbf{v} := \lambda\mu\mathbf{v}$
- **Distributive law**:
  - Vector $\rightarrow \lambda(\mathbf{v}+\mathbf{w}) = \lambda\mathbf{v}+\lambda\mathbf{w}$
  - Scalar $\rightarrow (\lambda+\mu)\mathbf{v} = \lambda\mathbf{v}+\mu\mathbf{v}$

Two vectors $\mathbf{a}$, $\mathbf{b}$ are parallel if there exists a scalar λ such that either $\mathbf{a}=\lambda\mathbf{b}$ or $\mathbf{b}=\lambda\mathbf{a}$.

Coordinates

The place coordinates on a plane the following must be specified:

- an origin point O
- a pair of vectors $\mathbf{i}$ and $\mathbf{j}$ that both have a unit length and are at right angles to each other (it is conventional to choose $\mathbf{j}$ at $\frac{\pi}{2}$) anticlockwise from $\mathbf{i}$

Every geometric vector $\mathbf{a}$ in the plane can be written as $\mathbf{a} = a_1\mathbf{i}+a_2\mathbf{j} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

1.2 Vector quantities and $\mathbb{R}^n$

Coordinate vectors can be generalised to be any number of components. Let n be a positive integer. An n-tuple or n-vector is an ordered list of n numbers $a_1, a_2, \ldots, a_n$ and can be either written as a row or column vector

The set of all n-tuples is denoted $\mathbb{R}^n$. Thus

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \in \mathbb{R}^2, \qquad \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \in \mathbb{R}^3, \qquad \begin{pmatrix} 0 \\ 1 \\ -1 \\ \pi \end{pmatrix} \in \mathbb{R}^4$$

1.2.1 Arithmetic on $\mathbf{R}^n$

Vector addition and scalar multiplication in $\mathbb{R}^n$ is defined coordinatewise, as previously demonstrated in $\mathbb{R}^2$ and $\mathbb{R}^3$.

$$\begin{pmatrix} a_1 \\ a_2 \\ \ldots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \ldots \\ b_n \end{pmatrix} := \begin{pmatrix} a_1+b_1 \\ a_2+b_2 \\ \ldots \\ a_n+b_n \end{pmatrix}$$

The 'zero element' is $\mathbf{0} = \begin{pmatrix} 0 \\ \dots \\ 0 \end{pmatrix}$, and the negative is given by $-\begin{pmatrix} a_1 \\ \dots \\ a_n \end{pmatrix} = \begin{pmatrix} -a_1 \\ \dots \\ -a_n \end{pmatrix}$

Note that each $\mathbb{R}^n$ is a separate system, eg. a 3-tuple cannot be added to a 7-tuple

The length of a vector $\mathbf{a}$ is defined as $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$

1.2.2 Lines in $\mathbb{R}^n$

A line L in $\mathbb{R}^n$ is determined by:

- a point A on the line, eg. with $\mathbf{a} = \overrightarrow{OA}$
- a direction, given by a non-zero vector v

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \quad \lambda \in \mathbb{R}$$

This form is referred to as parametric vector form of the line L.

A line in $\mathbb{R}^n$ is any set of the form

$$\{\mathbf{x} \in \mathbb{R}^n | \mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \quad \lambda \in \mathbb{R}, \mathbf{v} \neq \mathbf{0}, \mathbf{a} \in \mathbb{R}^n\}$$

1.2.3 Parametric to Cartesian form

Example

Write the line $\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\lambda \in \mathbb{R}$ in Cartesian form

$$\text{Let } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 + 2\lambda \\ -1 + \lambda \end{pmatrix}$$
$$\lambda = \frac{x-1}{2} = y + 1$$

Example

Consider $\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\lambda \in \mathbb{R}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 + \lambda \end{pmatrix}$$

In this case, $x = 1$.

1.2.4 Cartesian form for lines and planes in $\mathbb{R}^3$

A plane in xyz-space can be described by the equation

$$ax + by + cz = d$$

where not all a, b, c are 0. This is called the Cartesian form for the plane. The Cartesian form for a line in $\mathbb{R}^n$ can be defined as the intersection of two planes ($P_1 \cap P_2$). Usually, this can be written in the form

$$\frac{x-a_1}{v_1} = \frac{y-a_2}{v_2} = \frac{z-a_3}{v_3}$$

for some constants a_i, v_i.

Example

Find the parametric form for $\frac{x-2}{3} = \frac{y+1}{6} = \frac{z-3}{-2}$

$$\lambda = \frac{x-2}{3} = \frac{y+1}{6} = \frac{z-3}{-2}$$
$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3\lambda+2 \\ 6\lambda_1 \\ -2\lambda+3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

1.2.5 Parametric to Cartesian form for lines in $\mathbb{R}^3$

Example

Find a Cartesian form for $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, $\lambda \in \mathbb{R}$

$$\begin{pmatrix} 4\lambda+1 \\ 5\lambda+2 \\ 6\lambda+3 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\lambda = \frac{x-1}{4} = \frac{y-2}{5} = \frac{z-3}{6}$$

Example

Consider $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix}$

$$\lambda = \frac{x-1}{4} = \frac{y-2}{5}, \quad z = 3$$

1.2.6 Linear combination

Suppose that $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k} \in \mathbb{R}^n$. A linear combination of these vectors is a vector of the form

$$\lambda_1\mathbf{v_1} + \lambda_2\mathbf{v_2} + \cdots + \lambda_k\mathbf{v_k}, \quad \lambda_1, \ldots, \lambda_k \in \mathbb{R}$$

1.2.7 Span

The span of $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}$ is the set of all linear combinations of $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}$, ie.

$$\operatorname{span}(\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}) = \{\lambda_1\mathbf{v_1}, \lambda_2\mathbf{v_2}, \ldots, \lambda_k\mathbf{v_k} | \lambda_1, \ldots, \lambda_k \in \mathbb{R}\}$$

Note that $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$

1.2.8 Planes in $\mathbb{R}^n$

A plane in $\mathbb{R}^n$ is defined to be a set of the form

$$S = \{\mathbf{a} + \lambda_1\mathbf{v_1} + \lambda_2\mathbf{v_2} | \lambda_1, \lambda_2 \in \mathbb{R}\}$$

where $\mathbf{a}$, $\mathbf{v_1}$ and $\mathbf{v_2}$ are fixed vectors in $\mathbb{R}^n$, and $\mathbf{v_1}$ and $\mathbf{v_2}$ are not parallel.

The expression $\mathbf{x} = \mathbf{a} + \lambda_1\mathbf{v_1} + \lambda_2\mathbf{v_2}$, $\lambda_1, \lambda_2 \in \mathbb{R}$ is a parametric vector form for the plane through $\mathbf{a}$ parallel to the vectors $\mathbf{v_1}$ and $\mathbf{v_2}$.

Example

Find a parametric vector equation for the plane containing the point $\mathbf{a} = \begin{pmatrix}1\\2\\1\end{pmatrix}$, and line $\mathbf{x} = \begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}3\\-4\\2\end{pmatrix}$, $\lambda \in \mathbb{R}$.

$$\text{Let direction } \mathbf{v_2} = \begin{pmatrix}1\\2\\1\end{pmatrix} - \begin{pmatrix}0\\1\\2\end{pmatrix} = \begin{pmatrix}1\\1\\-1\end{pmatrix}$$

$$\mathbf{x} = \begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda_1\begin{pmatrix}3\\-4\\2\end{pmatrix} + \lambda_2\begin{pmatrix}1\\1\\-1\end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}$$

Example

Find the Cartesian equation of the plane in $\mathbb{R}^3$: $\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda_1\begin{pmatrix}-1\\1\\-1\end{pmatrix} + \lambda_2\begin{pmatrix}3\\2\\1\end{pmatrix}$, $\lambda_1, \lambda_2 \in \mathbb{R}$.

Equating coordinates and adding:

$$x + y = 3 + 5\lambda_2$$
$$y + z = 5 + 3\lambda_2$$

Hence $3(x+y) - 5(y+z) = 3(3) = 5(5) = -16$

$$\therefore -16 = 3x - 2y - 5z$$

Chapter 2

Vector Geometry

2.1 Angles via cosine rule

Let $\mathbf{a} = \overrightarrow{OA}, \mathbf{b} = \overrightarrow{OB} \in \mathbb{R}^n$ be non-zero. The cosine rule is as follows

$$|\mathbf{b} - \mathbf{a}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta$$

The formula for the length of a vector gives

$$\sum_i (b_i - a_i)^2 = \sum_i a_i^2 + \sum_i b_i^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta$$

Cancelling gives $-2|\mathbf{a}||\mathbf{b}|\cos\theta = \sum_i -2a_i b_i$ so

$$\cos\theta = \frac{\sum_i a_i b_i}{|\mathbf{a}||\mathbf{b}|}$$

θ is solvable as long as RHS lies in $[-1, 1]$ (Cauchy-Schwarz theorem)

The dot product of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is defined to be

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$$

Hence, the angle between two non-zero vectors $\mathbf{a} = \overrightarrow{OA}, \mathbf{b} = \overrightarrow{OB} \in \mathbb{R}^n$ is

$$\theta = \cos^{-1}\left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}\right)$$

Properties of the dot product

Suppose $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$. Then

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- $\mathbf{a} \cdot (\lambda \mathbf{b}) = (\lambda \mathbf{a}) \cdot \mathbf{b} = \lambda(\mathbf{a} \cdot \mathbf{b})$
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2 \geq 0$

2.1.1 Cauchy-Schwarz theorem

The Cauchy-Schwarz theorem is as follows

$$-|\mathbf{a}||\mathbf{b}| \leq \mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}||\mathbf{b}|$$

Proof

The inequality holds when $\mathbf{b} = \mathbf{0}$ so it is assumed $\mathbf{b} \neq \mathbf{0}$. Consider the real function of (the real variable) λ

$$\begin{aligned} q(\lambda) &= |\mathbf{a} = \lambda \mathbf{b}|^2 \geq 0 \\ q(\lambda) &= (\mathbf{a} - \lambda \mathbf{b}) \cdot (\mathbf{a} - \lambda \mathbf{b}) \\ &= |\mathbf{a}|^2 - 2\lambda \mathbf{a} \cdot \mathbf{b} + \lambda^2 |\mathbf{b}|^2 \end{aligned}$$

The discriminant of this quadratic function of λ must be non-positive, hence

$$0 \geq (-2\mathbf{a} \cdot \mathbf{b})^2 - 4|\mathbf{a}|^2|\mathbf{b}|^2 = 4(\mathbf{a} \cdot \mathbf{b})^2 - 4|\mathbf{a}|^2|\mathbf{b}|^2 \implies (\mathbf{a} \cdot \mathbf{b})^2 \leq |\mathbf{a}|^2|\mathbf{b}|^2$$

2.2 Orthogonality

Two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are said to be orthogonal if $\mathbf{a} \cdot \mathbf{b} = 0$, ie. the angle between them is $\frac{\pi}{2}$

Example

Consider $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 1 \\ 1 \\ 2 \end{pmatrix}$

$$\mathbf{a} \cdot \mathbf{b} = (1 \times -2) + (1 \times 1) + (-1 \times 1) + (1 \times 2) = 0$$

therefore, $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

If $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are orthogonal then

$$|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2$$

2.2.1 Orthonormal sets of vectors

The vectors $\mathbf{v_1}, \dots, \mathbf{v_k} \in \mathbb{R}^n$ form an orthogonal set if they are mutually orthogonal. Furthermore, if they all have length 1, the vectors are orthonormal.

Equivalently, $\mathbf{v_1}, \dots, \mathbf{v_k}$ are orthonormal if

$$v_i \cdot v_j = \delta_{ij} := \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

[δ_{ij} is called the Kronecker delta symbol]

Arithmetic prperties of the cross product

For $\mathbf{a},\mathbf{b},\mathbf{c} \in \mathbb{R}$ and $\lambda \in \mathbb{R}$

- $\mathbf{a} \times \mathbf{a} = 0$
- $\times$ is distributive:
  - $\mathbf{a} \times (\mathbf{b}+\mathbf{c}) = (\mathbf{a}\times\mathbf{b}) + (\mathbf{a}\times\mathbf{c})$
  - $(\mathbf{a}+\mathbf{b}) \times \mathbf{c} = (\mathbf{a}\times\mathbf{c}) + (\mathbf{b}\times\mathbf{c})$
- $(\lambda\mathbf{a}) \times \mathbf{b} = \lambda(\mathbf{a}\times\mathbf{v}) = \mathbf{a}\times(\lambda\mathbf{b})$
- Since $\mathbf{a}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$, $\times$ is not commutative
- The cross product is not associative

The magnitude of $\mathbf{a}\times\mathbf{b}$

Suppose that $\mathbf{a},\mathbf{b} \in \mathbb{R}^3$ are nonzero, with angle θ between them. Then $|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$. Since $\theta \in [0,\pi]$ in $\mathbb{R}^3$, therefore $\sin\theta \geq 0$. Thus, it is sufficient that

$$|\mathbf{a}\times\mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2\sin^2\theta$$

Areas of parallelgroms via cross product

The area of a parallelogram is

$$A = \text{base} \times \text{perp height} = |\mathbf{a}||\mathbf{b}|\sin\theta$$

Direction of $\mathbf{a}\times\mathbf{b}$

Row swapping of determinants gives the following

- $\mathbf{a}\dot{(}\mathbf{b}\times\mathbf{c}) = -\mathbf{a}\dot{(}\mathbf{c}\times\mathbf{b}) = \mathbf{c}\dot{(}\mathbf{a}\times\mathbf{b})$
- $\mathbf{a}\dot{(}\mathbf{a}\times\mathbf{b}) = 0 = \mathbf{a}\dot{(}\mathbf{b}\times\mathbf{b})$

The vector $\mathbf{a}\times\mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$

Parametric to Cartesian form via point-normal

Example

Find a point-normal, and hence a Cartesian form for the plane $\mathbf{x} = \begin{pmatrix}-3\\1\\2\end{pmatrix} + \lambda_1\begin{pmatrix}1\\1\\-2\end{pmatrix} + \lambda_2\begin{pmatrix}4\\0\\3\end{pmatrix}, \quad \lambda_1,\lambda_2 \in \mathbb{R}$

A point on the line is $\mathbf{a} = \begin{pmatrix}-3\\1\\2\end{pmatrix}$

A vector normal to the plane is $\mathbf{n} = \begin{pmatrix}1\\1\\-2\end{pmatrix} \times \begin{pmatrix}4\\0\\3\end{pmatrix}$

Chapter 3

Complex Numbers

3.0.1 Fields

A field is a non empty set for which a rule for addition and multiplication are defined. The field must define values for the following axioms

1. **Closure under Addition.** If $x, y \in \mathbb{F}$ then $x + y \in \mathbb{F}$.
2. **Associative Law of Addition.** $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}$.
3. **Commutative Law of Addition.** $x + y = y + x$ for all $x, y \in \mathbb{F}$.
4. **Existence of a Zero.** There exists an element of $\mathbb{F}$ (usually written as 0) such that $0 + x = x + 0 = x$ for all $x \in \mathbb{F}$.
5. **Existence of a Negative.** For each $x \in \mathbb{F}$, there exists an element $w \in \mathbb{F}$ (usually written as $-x$) such that x + w = w + x = 0.
6. **Closure under Multiplication.** If $x, y \in \mathbb{F}$ then $xy \in \mathbb{F}$.
7. **Associative Law of Multiplication.** $x(yz) = (xy)z$ for all $x, y, z \in \mathbb{F}$.
8. **Commutative Law of Multiplication.** $xy = yx$ for all $x, y \in \mathbb{F}$.
9. **Existence of a One.** There exists a non-zero element of $\mathbb{F}$ (usually written as 1) such that $x1 = 1x = x$ for all $x \in \mathbb{F}$.
10. **Existence of an Inverse for Multiplication.** For each non-zero $x \in \mathbb{F}$, there exists an element w of $\mathbb{F}$ (usually written as $1/x$ or x^{-1}) such that $xw = wx = 1$.
11. **Distributive Law.** $x(y + z) = xy + xz$ for all $x, y, z \in \mathbb{F}$.
12. **Distributive Law.** $(x + y)z = xz + yz$ for all $x, y, z \in \mathbb{F}$.

3.0.2 Complex numbers

A complex number is a formal expression of the form $a + bi$ for some $a, b \in \mathbb{R}$. In particular, two such numbers $a + bi$, $a' + b'i$ are equal iff $a = a', b = b'$ as real numbers. The real part of $a + bi$ is $\mathrm{Re}(a + bi) = a$ and the imaginary part is $\mathrm{Im}(a + bi) = b$.

Arithmetic of complex numbers

Given the complex numbers $a + bi, a' + b'i$, addition and multiplication is defined as follows

$$\begin{aligned}(a+bi)+(a'+b'i) &= (a+a')+(b+b')i \\ (a+bi)(a'+b'i) &= (aa'-bb')+(ab'+a'b)i\end{aligned}$$

The complex set $\mathbb{C}$ extends the real number system since complex numbers of form $a+0i$ add and multiply the same way as real numbers. It contains the number i satisfying $i^2 = -1$

Cartesian form

A complex number z written in the form $a+bi$ with $a, b \in \mathbb{R}$ is called the Cartesian form

Example

Express $\frac{1+i}{1-i} - \frac{1-i}{1+i}$ in Cartesian form

$$\begin{aligned}\frac{1+i}{1-i} - \frac{1-i}{1+i} &= \frac{(1+i)^2-(1-i)^2}{(1-i)(1+i)} \\ &= \frac{(1+2i+i^2)-(1-2i+i^2)}{1^2+1^2} \\ &= \frac{4i}{2} \\ &= 2i\end{aligned}$$

Conjugation

The conjugate of a complex number $z = a+bi$ is defined as $\overline{z} = a - bi$

- z is real iff $\overline{z} = z$
- $\overline{\overline{z}} = z$
- $\overline{z+w} = \overline{z} + \overline{w}$ and $\overline{z-w} = \overline{z} - \overline{w}$
- $\overline{zw} = \overline{z}\,\overline{w}$ and $\overline{\frac{z}{w}} = \frac{\overline{x}}{\overline{w}}$
- $\mathrm{Re}(z) = \frac{1}{2}(z+\overline{z})$ and $\mathrm{Im}(z) = \frac{1}{2i}(z-\overline{z})$

Example

Show that for any $z \in \mathbb{C}$, $(i+5)z - (i-5)\overline{z}$ is real.

$$\begin{aligned}\text{Let } w &= (i+5)z-(i-5)\overline{z} \\ \overline{w} &= \overline{(i+5)z-(i-5)\overline{z}} \\ &= (5-i)\overline{z}-(-i-5)z \\ &= (i-5)\overline{z}+(i+5)z \\ &= w\end{aligned}$$

3.1 The Argand diagram

Similarly to how real numbers can be represented on a plane, complex numbers can be represented on the complex plane (the Argand diagram).

$z = a + bi$ is represented by the point with coords $(a,b) = (\mathrm{Re}(z), \mathrm{Im}(z))$

Adding complex numbers by adding real and imaginary parts (ie. coordinatewise) can be represented geometrically by the addition of vectors.

3.1.1 Polar form

Given the Cartesian form of a complex number $z = x + iy$, $x, y \in \mathbb{R}$, polar coordinates for $z = (r, \theta)$ can be found

$$z = r\cos\theta + (r\sin\theta)i$$

Let $z = x + iy$, $x, y \in \mathbb{R}$

- The **modulus** of z is defined as $|z| = r = \sqrt{x^2 + y^2}$ so $z\bar{z} = |z|^2$
- If $z \neq 0$, an **argument** for z is any $\theta = \arg z$ so that $\tan\theta = \frac{y}{x}$
- Furthermore, $\theta =: \mathrm{Arg}\, z$ is the **principal argument** if $\theta \in (-\pi, \pi]$

Example

Find the modulus and principal argument of $-1 - \sqrt{3}i$.

$$\begin{aligned} |-1-\sqrt{3}i| &= \sqrt{1^2 + (\sqrt{3})^2} = 2 \\ \arctan\frac{-\sqrt{3}}{-1} &= \arctan\sqrt{3} = \frac{\pi}{3} \\ \mathrm{Arg}(-1-\sqrt{3}) &= \frac{\pi}{3} - \pi = -\frac{2\pi}{3} \end{aligned}$$

3.1.2 Euler's formula

For $\theta \in \mathbb{R}$, it is defined $e^{i\theta} = \cos\theta + i\sin\theta$. This formula is reasonable by

- $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$
- (De Moivre's theorem) For $n \in \mathbb{Z}$, $\left(e^{i\theta}\right)^n = e^{in\theta}$
- $\frac{d}{d\theta}(e^{i\theta}) = ie^{i\theta}$

Arithmetic of polar forms

The polar form of z is $z = re^{i\theta}$ where $r = |z|$ and θ is an argument of z. The above formulae give

$$r_1e^{i\theta_1}r_2e^{i\theta_2} = (r_1r_2)e^{i(\theta_1+\theta_2)} \quad , \quad \left(re^{i\theta}\right)^{-1} = r^{-1}e^{-i\theta}$$

ie. multiplication of complex numbers can be achieved by multiplying the moduli and adding the arguments. Inverting a complex number inverts the modulus and negates the argument.

Example

Find the exact value of $\operatorname{Arg}\frac{1+i}{1+\sqrt{3}i}$

$$\begin{aligned}\arg\frac{1+i}{1+\sqrt{3}i} &= \arg(1+i)-\arg(1+\sqrt{3}i)\\ &= \frac{\pi}{4}-\frac{\pi}{3}\\ \operatorname{Arg}\frac{1+i}{1+\sqrt{3}i} &= -\frac{\pi}{12}\end{aligned}$$

Example

Let $z \in \mathbb{C}$ have $|z| = 1$. Show that $w = \frac{i-z}{i+z}$ is purely imaginary in the sense that $\operatorname{Re}(w) = 0$ Interpret the result geometrically.

$$\operatorname{Re}(w) = \frac{1}{2}(w+\overline{w})$$

$$\begin{aligned}\overline{w} &= \frac{\overline{i-z}}{\overline{i+z}}\\ &= \frac{-i-\overline{z}}{-i+\overline{z}} \times \frac{iz}{iz}\\ &= \frac{z-i|z|^2}{z+i}\\ &= -w\end{aligned}$$

Therefore, w is purely imaginary

Example

Find the complex sqaure roots $\pm z$ of $16-30i$

Let $z = a+bi$, $a,b \in \mathbb{R}$ such that $z^2 = 16-30i$

$$\begin{aligned}&\therefore 16-30i = (a+bi)^2 = a^2+2abi-b^2\\ &= a^2-b^2+2abi\\ &\text{Real part gives: } 16 = a^2-b^2\\ &\text{Imaginary part gives: } -30 = 2ab \Leftrightarrow -15 = ab\end{aligned}$$

$$z = \pm(5-3i)$$

Example

Solve $z^2+(1-i)z+(-4+8i)=0$

$$
\begin{aligned}
z &= \frac{-(1-i)\pm\sqrt{(1-i)^2-4(1)(-4+8i)}}{2(1)} \\
\Delta &= b^2-4ac=(1-i)^2-4(1)(-4+8i)=1^2+2i+i^2+16-32i=16-30i \\
&\text{From above, } \sqrt{\Delta}=\pm(5-3i) \\
z &= \frac{-(1+i)\pm(5-3i)}{2}=\frac{4-4i}{2} \text{ or } \frac{-6+2i}{2} \\
&= 2-2i,\ -3+i
\end{aligned}
$$

3.1.3 Powers of complex numbers

Example

Find $w=(1+i)^{18}$

$$
\begin{aligned}
\text{Let } z &= 1+i=\sqrt{2}e^{i\frac{\pi}{4}} \\
w &= z^{18} \\
&= (\sqrt{2}e^{i\frac{\pi}{4}})^{18} \\
&= 2^9 e^{i\frac{9\pi}{2}} \\
&= 2^9 e^{i\frac{\pi}{2}} \\
&= 2^9 i
\end{aligned}
$$

3.1.4 Roots of complex numbers

Example

Solve $z^4=i$.

Solve in polar form for $z=re^{i\theta}$, $\pi<\theta\leq\pi$

Equate moduli: $1=|i|=|z^4|=|z|^4$, $\therefore r=1$

Equate arguments: $\arg(z^4)=\arg i$

$4\ \mathrm{Arg}(z)=\frac{\pi}{2}$

...(got lazy, answer trivial)

3.1.5 Expressing trigonometric polynomials as polynomials in $\cos\theta$, $\sin\theta$

A trigonometric polynomial is a linear combination of functions of the form $\cos n\theta$, $\sin n\theta$.

Example

Use De Moivre's theorem to show $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$

$$\begin{aligned} cos3\theta &= \mathrm{Re}(e^{i3\theta}) \\ &= \mathrm{Re}(\cos\theta + i\sin\theta)^3 \\ &= Re(\cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3 theta) \\ &= \cos^3\theta - 3\cos\theta\sin^2\theta \\ &= \cos^3\theta - 3\cos\theta(1-\cos^2\theta) \\ &= 4\cos^3\theta - 3\cos\theta \end{aligned}$$